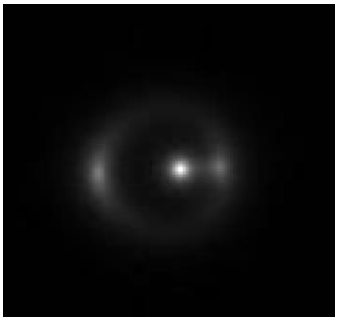
Google Summer of Code 2026
Machine Learning for Science (ML4SCI)



GSoC 2026

Project Proposal

By

Sourish Dutta

On

Gravitational Lens Finding (DeepLense)

Google Summer of Code 2026 | Project Proposal

Gravitational Lens Finding

DeepLense | ML4SCI

Submitted by

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IIT (ISM) Dhanbad | B.Tech (2025–2029)

CyberLabs - ML Division | IST (UTC+5:30)

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Table of Contents

About Me **4**
Student Information 4
University Information 4
Background Work and Programming Skills **4**
Evaluation Task **6**
Common Test I - Multi-Class Substructure Classification 6
Understanding the Data 6
My Approach and Implementation 7
Results 7
Specific Test V - Gravitational Lens Finding 9
Understanding the Data 9
My Approach and Implementation 9
Results 9
Project Information **12**
Abstract **12**
Ideas **12**
Approach **14**
1. Establish a Strong Baseline 14
2. Implement Domain Adaptation 14
3. Run the Real-Data Lens Search 15
4. Cross-Survey Evaluation 15
Timeline (Tentative) **16**
Pre-GSoC Period (until 1 June) 16
Deliverables **19**
Future Scope **19**
The Motivation for GSoC **19**
Achievements **20**
Hackathons and Competitions 20
Relevant Projects 21
Certifications 21

About Me

Student Information

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University Information

University	Indian Institute of Technology (ISM) Dhanbad
Department	Petroleum Engineering
Degree	B.Tech
Batch	2025 - 2029
ML Club	CyberLabs - ML Division (Member since January 2026)

Background Work and Programming Skills

I study at IIT (ISM) Dhanbad, one of India's top technical institutes. My engineering coursework focuses on applied physics, including reservoir simulation, seismic inversion, and subsurface data analysis. These problems share similar methods with gravitational lensing. They involve a physical forward model, noisy high-dimensional observations, and the goal of inverting or classifying them. This approach feels natural to me.

Before university, I was already creating ML systems from scratch. I built a scalar autograd engine with a dynamic computation graph and manual backpropagation. I also developed a character-level language model with batch normalization and Kaiming initialization, as well as feedforward networks in NumPy. I joined IIT ISM's top coding club, CyberLabs, because I was already engaged in this work.

My Task V result demonstrates my research readiness. I achieved a test AUC of 0.9502 on real observational lens-finding data. I did this by accurately diagnosing and addressing a 100:1 class imbalance meticulously.

- **Languages:** I am proficient with Python (primary), C/C++ for algorithms and systems work
- **PyTorch:** custom training loops, mixed precision (torch.cuda.amp), DataLoaders, WeightedRandomSampler, checkpoint management, AMP API migration
- **Deep Learning architectures:** CNNs including ResNet18, EfficientNet-B0/B3, MobileNetV3; two-phase fine-tuning; ImageNet transfer learning
- **Training infrastructure:** Kaggle dual T4 GPU environments, resume-from-checkpoint workflows, OOM debugging
- **Scientific Python:** NumPy, Pandas, Matplotlib, Scikit-learn, ROC/AUC evaluation, class imbalance metrics
- **EEG and signal processing:** MNE-Python, CEBRA, ICA, pyprep (used for the NeuroDyads GSoC pre-task)
- **Other tools:** Flask, Next.js, FastAPI, QuTiP, genetic algorithms, TF-IDF retrieval

Evaluation Tasks

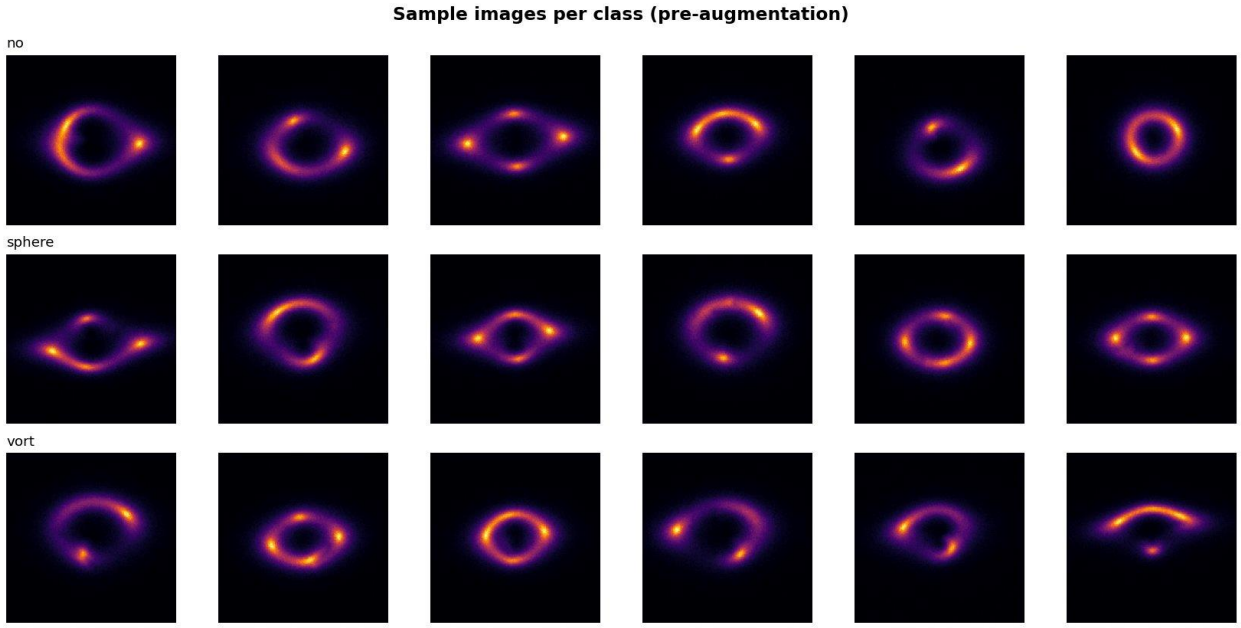
I completed both the mandatory Common Test I (multi-class classification) and Specific Test V (gravitational lens finding), which is the required test for this project track. Below I describe my approach, implementation choices, and results for each.

Common Test I - Multi-Class Substructure Classification

Understanding the Data

The task is three-class classification of simulated strong lensing images: no substructure, subhalo substructure and vortex substructure (a signature of superfluid dark matter condensates). Images are single-channel, 150×150 pixels, normalised using min-max scaling. The dataset has mild class imbalance - the sphere class is underrepresented, which causes naive classifiers to under-recall it.

The scientific context comes directly from Alexander et al. (2019): different dark matter models produce morphologically distinct signatures in strong lensing images, and CNNs can distinguish these more reliably than traditional statistical methods. The vortex class is particularly compelling - detecting it would be direct evidence for superfluid dark matter, outside the standard WIMP paradigm.



The Given Dataset

My Approach and Implementation

Model: I used EfficientNet-B0 with ImageNet-pretrained weights. I adapted the first convolutional layer to accept single-channel input by averaging the pretrained RGB weights across the channel dimension. I replaced the classifier head with Linear (1280 → 3) for three-class output.

Training strategy: I used a two-phase approach. In Phase 1 (3 epochs) I froze all backbone layers and trained only the classifier head (LR = 1e-3) to initialise it before back-propagating into the backbone. In Phase 2 (12 epochs) I unfroze everything and trained with LR = 1e-4. This prevents destroying the pretrained feature representations before the head has learned a reasonable mapping.

Class imbalance: I used WeightedRandomSampler to ensure balanced batches. I also tested focal loss and class-weighted cross-entropy; the combination of WeightedRandomSampler with standard cross-entropy gave the best macro-AUC in practice.

Augmentation: Random horizontal and vertical flips, random rotation (±30°), and random affine jitter - restricted to physically invariant transformations.

Optimisation: AdamW (weight_decay=1e-4), cosine annealing LR schedule, mixed precision. Best checkpoint saved by validation macro-AUC, not loss.

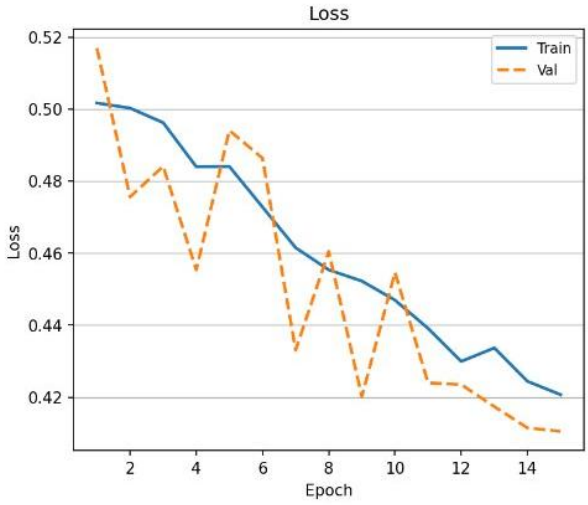
Results

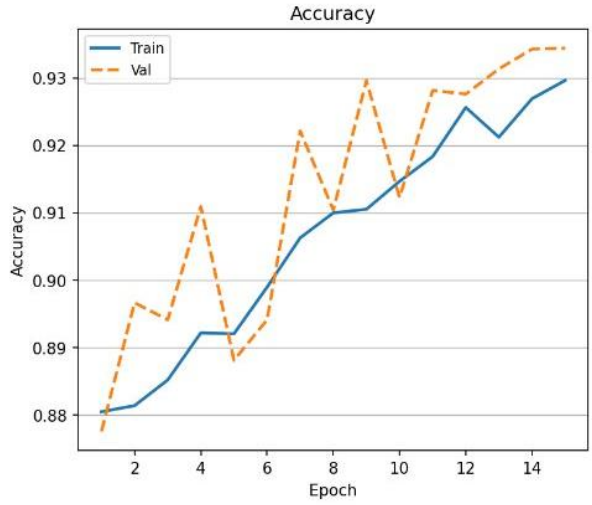
Metric	Value
Macro AUC (best checkpoint, epoch 15)	**0.9893**
Overall Accuracy	**93.44%**
AUC - No Substructure class	**0.993**
AUC - Sphere Substructure class	**0.982**
AUC - Vortex Substructure class	**0.993**
Sphere class recall (confusion matrix)	**86% (2162 / 2500)**
Vortex class recall (confusion matrix)	**95% (2375 / 2500)**

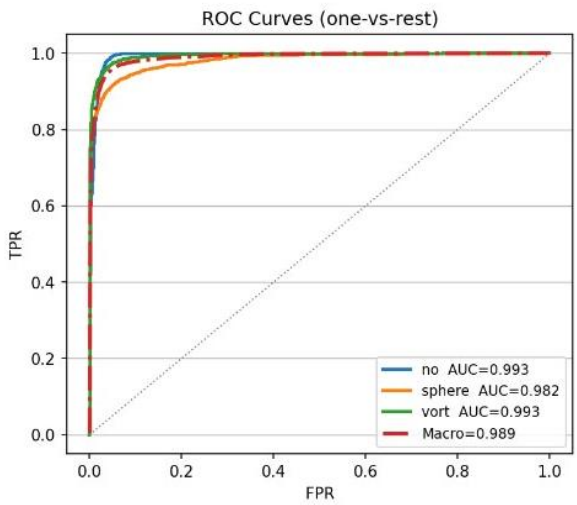
Key observation: The sphere class had the lowest recall throughout training (86%). This is physically expected - spherical subhalos produce diffuse, extended lensing signatures that partially overlap in feature space with the no-substructure class. Alexander et al. (2019) also reported the sphere class as the hardest to classify in their ResNet benchmarks.

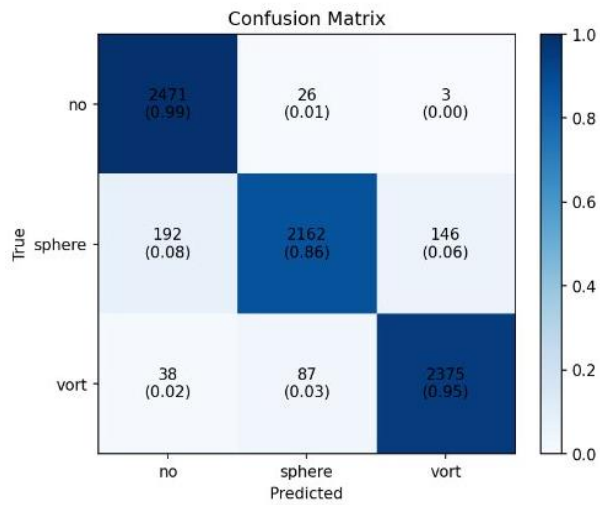
Code: https://github.com/sourishkumardutta/GSoC_DeepLense_PreTask/tree/main/Common_Task

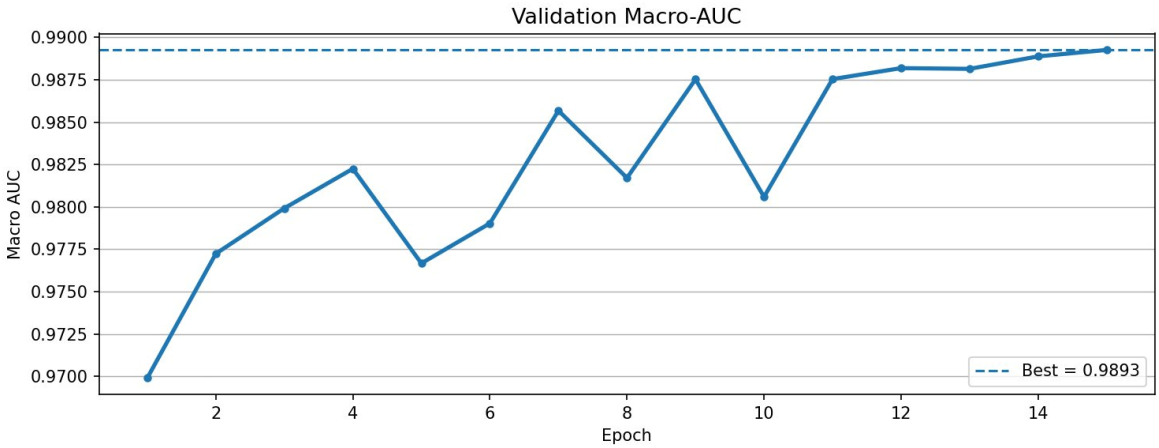
Plots


Loss
Train
Val
Loss
Epoch


Accuracy
Train
Val
Accuracy
Epoch


ROC Curves (one-vs-rest)
no AUC=0.993
sphere AUC=0.982
vort AUC=0.993
Macro=0.989
TPR
FPR


Confusion Matrix
no
sphere
vort
2471 (0.99)
26 (0.01)
3 (0.00)
192 (0.08)
2162 (0.86)
146 (0.06)
38 (0.02)
87 (0.03)
2375 (0.95)
True
Predicted


Validation Macro-AUC
Best = 0.9893
Macro AUC
Epoch

Specific Test V - Gravitational Lens Finding

Understanding the Data

This task uses real observational data - galaxy images structured as train_lenses / train_nonlenses and test_lenses / test_nonlenses. Each object is a 3-channel (g, r, i filter) image of shape (3, 64, 64). The defining challenge is severe class imbalance: roughly 100× more non-lenses than lenses. This is not just a machine learning nuisance - it reflects the fundamental reality of automated lens searches. In real surveys, lenses occur at roughly 1 per 1,000 - 10,000 objects, so even a 1% false positive rate generates an overwhelming number of false positives that human reviewers cannot practically process. Understanding this drove every choice I made in my solution.

My Approach and Implementation

Model: ResNet18 pretrained on ImageNet. The three input channels map naturally to the three survey filters (g, r, i). I replaced the final fully connected layer with Linear (512 → 1) for binary output.

Loss function: BCEWithLogitsLoss with a positive class weight set to the empirical non-lens to lens ratio (approximately 100). This directly penalises false negatives more heavily during training - appropriate given the scientific priority of not missing real lenses.

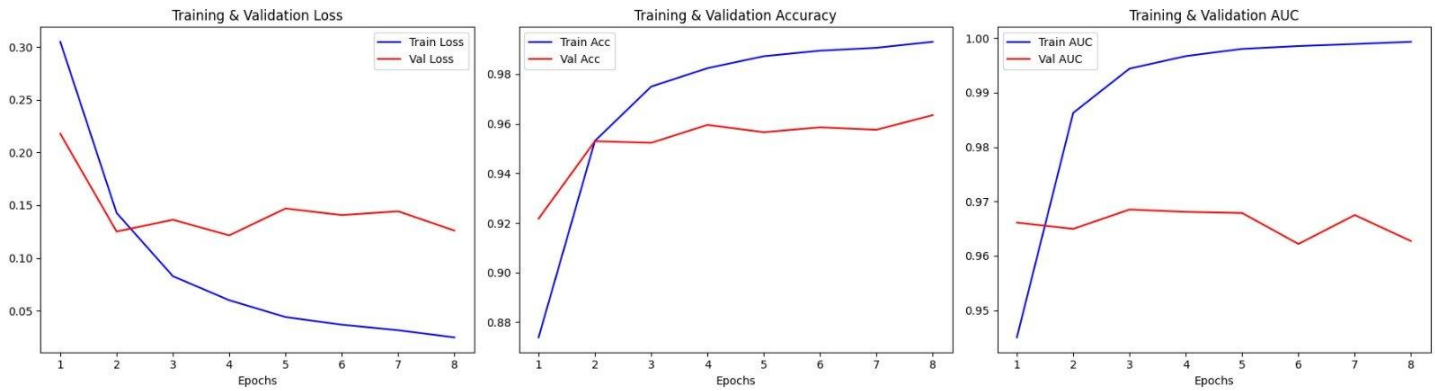
Sampling: WeightedRandomSampler to create roughly balanced batches, combined with augmentation (random flips, rotations, crops) applied specifically to the minority lens class.

Threshold selection: I chose the decision threshold on the validation set to maximise F1, rather than using the default 0.5. In highly imbalanced settings the default threshold is almost never optimal.

Training: 8 epochs, AdamW optimiser, cosine annealing schedule, mixed precision.

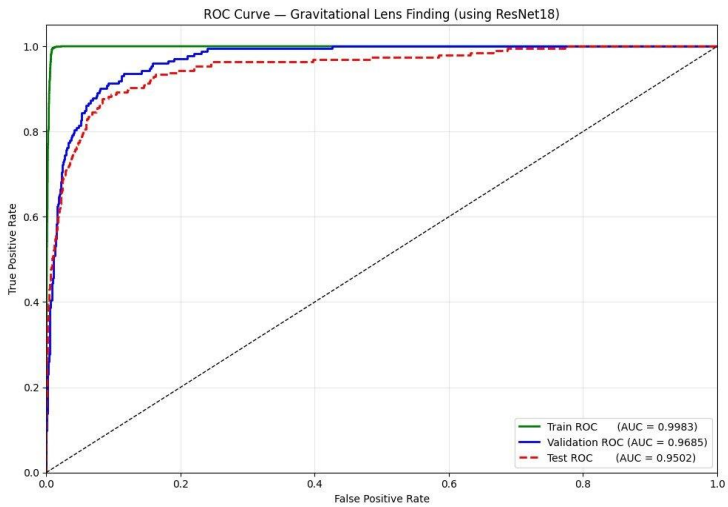
Results

Metric	Value
Train AUC	**0.9983**
Validation AUC	**0.9685**
Test AUC	**0.9502**
True Positives (lenses correctly identified)	**143 / 195 (73.3%)**
False Positives (non-lenses above threshold)	**692 / 19,455 (3.6%)**
True Negatives (non-lenses correctly rejected)	**18,763 / 19,455 (96.4%)**

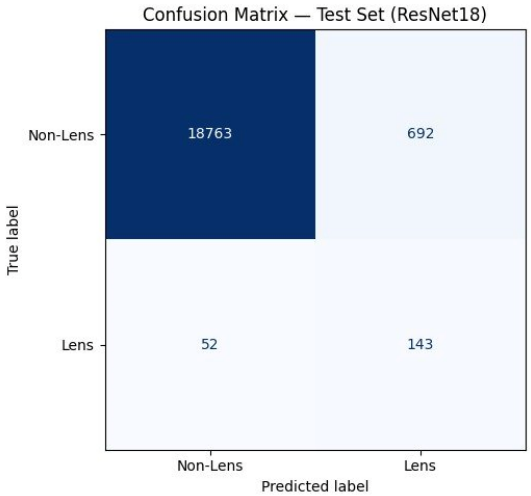


ResNet18 training curves for Task V: loss, accuracy, and AUC across 8 epochs.

Validation AUC stabilises around 0.965–0.970 from epoch 4 onward.



Train AUC = 0.9983, Validation AUC = 0.9685, Test AUC = 0.9502.



143 of 195 lenses correctly identified, 692 false positives out of 19,455 non-lenses.

Analysis:

The drop from train AUC (0.9983) to test AUC (0.9502) is the central diagnosis of this project, not a failure. It reflects the domain gap between what the model learns on a limited training sample and what it encounters across the full test set.

For context: Parul et al. (2024) showed that a naive simulation-trained ENN classifier drops from AUROC 0.995 on simulated data to 0.921 on real HSC observations. My **ResNet18 baseline at 0.**9502 - without any domain adaptation - **already exceeds their unadapted real-data baseline.** That gives me a solid foundation for the domain adaptation work I plan to do during GSoC.

Code: https://github.com/sourishkumardutta/GSoC_DeepLense_PreTask/tree/main/Specific_Test_V

Project Information

Organisation	ML4SCI
Project Title	Gravitational Lens Finding
Sub-project	DeepLense
Duration	175 hours (medium-length project)
Difficulty	Intermediate / Advanced
Mentor 1	Sergei Gleyzer - University of Alabama
Mentor 2	Michael Toomey - Massachusetts Institute of Technology
Mentor 3	Anna Parul - Observatoire de Paris
Reference papers	Alexander et al. 2019 (arXiv:1909.07346); Parul et al. NeurIPS ML4PS 2024

Abstract

Upcoming sky surveys like the Vera Rubin Observatory's LSST and Euclid are expected to find about 100,000 strong gravitational lenses. This is a huge increase from the few thousand we know about today. The main thing is not in finding faint objects but in achieving precision. Non-lenses outnumber lenses by a thousand to one or more. Classifiers trained on simulated data often struggle with real observations.

Parul et al. (NeurIPS 2024) quantified this issue. A model that scored 0.995 AUROC on simulations fell to 0.921 on real HSC-SSP images. This project builds on that finding.

The plan is to:

a) Train and benchmark CNN and equivariant network classifiers for lens detection, using TPR at FPR=0.01 along with AUC, which reflects what matters at survey scale

b) Implement and compare domain adaptation methods like ADDA, WDGRL, and supervised adaptation to bridge the gap between simulated training data and real observations

c) Run the resulting pipeline on real HSC-SSP PDR2 Wide data to find new lens candidates and analyze their properties

d) Test how well the pipeline works across different surveys.

The goal is to create a functioning, well-documented lens-finding pipeline that can support the automated discovery infrastructure needed for next-generation surveys.

Ideas

My plan for the summer is built around four interconnected ideas. They inform each other - the domain adaptation work only makes sense on top of a solid baseline, and the real-data search validates everything that came before it.

- **Equivariant architecture for lens finding:** Standard CNNs learn rotational invariance approximately, through augmentation. Gravitational lenses are rotationally symmetric by physics. I want to implement an ENN (equivariant neural network, e2cnn library, D4 symmetry group) and benchmark it against my ResNet18 baseline on both AUC and TPR. The NeurIPS 2024 results suggest it generalises better - this becomes the backbone for everything downstream if confirmed.

- **Domain adaptation across three methods:** I'll implement ADDA, WDGRL, and supervised domain adaptation (SDA), directly extending the Parul et al. (2024) benchmark. I specifically want to test whether adding spiral galaxies - the most common contaminant - to the SDA target set reproducibly reduces the false positive rate, as the NeurIPS paper found 3× improvement.

- **Real HSC-SSP PDR2 lens search:** I'll run the adapted model at scale on HSC-SSP PDR2 Wide data, generate a ranked candidate list, cross-match against known catalogues, and characterise top candidates. Anna Parul's group has the data infrastructure, and I plan to work closely with her on this phase.

- **Cross-survey generalisation:** I'll evaluate whether the HSC-adapted model degrades on a second survey dataset (DES or KiDS). This is directly listed on the project page and addresses a practically important question: does domain adaptation over-specialise to HSC image properties?

Approach

1. To Establish a Strong Baseline

My Task V ResNet18 result (test AUC 0.9502) is a starting point. The key insight from the NeurIPS 2024 paper is that Equivariant Neural Networks outperform standard CNNs on lens finding because they encode the physical symmetry of lensing directly into the architecture. I'll implement an ENN backbone using the e2cnn library with the dihedral group D4 - identity, rotations by 90° and 180°, horizontal and vertical reflections. This is the same backbone used by Parul et al.

Why does D4 matter?

A gravitational lens viewed at any rotation still belongs to the same class. A standard CNN learns this approximately by seeing many augmented examples. An ENN guarantees it by construction - the feature maps transform equivariantly under the group elements. This reduces the effective sample complexity and improves generalisation for the rare positive class where I have the fewest training examples.

Alongside the ENN I'll benchmark EfficientNet-B3 (already validated on the Common Task) and MobileNetV3-Small on the lens-finding dataset. I'll compare all architectures using both AUROC and TPR.

2. Implement Domain Adaptation

I'll implement all three domain adaptation methods from Parul et al. (2024) using the ENN backbone for consistency, allowing direct comparison to their published results.

ADDA (Adversarial Discriminative Domain Adaptation): I'll train a separate target encoder adversarially against a domain discriminator. The discriminator tries to distinguish source (simulated) from target (real) embeddings. The target encoder is trained to fool it. Once the embeddings are indistinguishable, the classifier - trained only on source data - can be applied to target embeddings with reduced domain shift.

WDGRL (Wasserstein Distance Guided Representation Learning): Instead of a binary discriminator, WDGRL uses Wasserstein distance to measure the gap between source and target representations. This provides more stable gradients, especially when the two distributions are initially far apart. I'll implement the three-component training loop: domain critic D, feature extractor M, and classifier C.

SDA (Supervised Domain Adaptation): Using the small set of known real lenses as labelled target data, SDA applies a semantic alignment loss (pulling same-class samples from source and target together) and a separation loss (pushing different-class samples apart). I'll also test explicitly adding spiral galaxies to the target set, replicating and extending Table 2 of the NeurIPS paper.

3. To Run the Real-Data Lens Search

Once I have a validated adapted model, I'll run it on the full HSC-SSP PDR2 Wide layer. This involves:

- **Extracting cutouts** (64×64) for all objects in the PDR2 Wide catalogue
- **Batched GPU inference, score assignment, and score distribution analysis** across approximately 10^7 objects
- **Threshold selection**: choosing the operating point that balances completeness versus manageable false positive counts for visual inspection
- **Cross-matching top candidates** against the Master Lens Database and SuGOHI catalogue to identify what is already known and what is new
- **False positive analysis**: identifying which contaminant types (spiral galaxies, ring galaxies, merging galaxies, imaging artefacts) dominate above the threshold - this directly informs where future model improvement effort should go

4. Cross-Survey Evaluation

After the **HSC search**, I'll evaluate the trained and adapted models on at least one additional dataset - either the DES strong lensing catalogue or the KiDS dataset - to measure how well the domain-adapted representations transfer beyond HSC.

This tests whether domain adaptation over-specialises to HSC image properties or learns more general features that carry over to other instruments.

Timeline (Tentative)

Total project length: 175 hours. Coding period: approximately 12 weeks at roughly 14–15 hours per week.

I don't have any major commitments in summer so I will have several weeks of preparation time before the coding period begins.

Pre-GSoC Period (until 1 June)

- Re-read and annotate Alexander et al. (2019) and Parul et al. (2024) in detail
- Reproduce the WDGRL training loop from scratch on a small toy dataset before Week 4 of the project, not during it
- Coordinate with mentors early to get HSC-SSP PDR2 data access credentials sorted
- Set up the project repository: folder structure, README, environment spec, CI/linting
- Run a dry-run comparison of ENN (e2cnn, D4) versus ResNet18 on the Task V dataset to validate my ENN implementation

Week	Dates (2026)	Tasks and Milestones
Phase 1 - Baseline Optimisation and Architecture Benchmarking (Weeks 1–3)		
W1	Jun 2–8	• Set up full data pipeline for HSC gri cutouts: normalisation, augmentation, DataLoader • Implement ENN backbone with e2cnn (D4 symmetry group) for binary lens classification • Initial comparison: ENN vs ResNet18 on Task V dataset - record AUROC and $TPR_{0.01}$
W2	Jun 9–15	• Benchmark EfficientNet-B3 and MobileNetV3-Small on the lens-finding task • Ablation: standard ImageNet augmentation vs physics-appropriate augmentation (360° rotation, PSF noise) • Hyperparameter sweep (LR, weight decay, scheduler) for top two architectures

W3	Jun 16–22	• Finalise best baseline architecture by $TPR_{0.01}$ and AUROC on validation set • Document Phase 1 results in repo: benchmark table and evaluation plots • Milestone: best baseline AUROC ≥ 0.96 on validation set
Phase 2 - Domain Adaptation Implementation (Weeks 4–6)		
W4	Jun 23–29	• Implement ADDA: supervised training of source ENN encoder on simulated lenses • Implement target encoder and domain discriminator; set up minimax training loop • Initial ADDA evaluation on real HSC test set: compare AUROC and $TPR_{0.01}$ vs unadapted baseline
W5	Jun 30–Jul 6	• Implement WDGRL: Wasserstein critic, feature extractor, and classifier • Direct comparison of ADDA vs WDGRL on the same held-out test set • Hyperparameter tuning for both unsupervised domain adaptation methods
W6	Jul 7–13	• Implement SDA: semantic alignment loss and separation loss • Test SDA with spiral galaxies added to target dataset (replicating Table 2 of NeurIPS 2024) • Write Phase 2 summary: full comparison table (AUROC, $TPR_{0.01}$, $F1_{0.01}$) for all three methods
Midterm Evaluation [~14–20 July, 2026] - Progress review with mentors, plan adjusted if needed		
Phase 3 - Real-Data Lens Search in HSC-SSP PDR2 Wide (Weeks 7–9)		
W7	Jul 21–27	• Set up HSC data loading pipeline (coordinate with Anna Parul's group on data access) • Run best adapted model on full PDR2 Wide catalogue using batched GPU inference • Score distribution analysis; identify candidate threshold range

W8	Jul 28–Aug 3	• Apply decision threshold; generate preliminary candidate list • Visual inspection of top 200 candidates with mentor input • Cross-match against Master Lens Database and SuGOHI catalogue
W9	Aug 4–10	• Characterise candidates: gri magnitudes, Einstein radius estimates, SDSS redshift cross-match • False positive analysis: identify dominant contaminant types above the threshold • Milestone: candidate catalogue with basic photometric characterisation, documented in repo
Phase 4 - Cross-Survey Evaluation, Extensions, and Wrap-up (Weeks 10–12)		
W10	Aug 11–17	• Cross-survey evaluation: apply HSC-trained model to DES or KiDS dataset • Analyse generalisation gap between HSC-adapted and native performance on second survey • Document cross-survey comparison
W11	Aug 18–24	• Buffer week: address any Phase 3–4 outstanding issues • Write comprehensive README with setup instructions and reproducibility guide • Create demo notebooks for each pipeline stage
W12	Aug 25–31	• Final code cleanup, documentation review, and repository polish • Write short project blog post for the ML4SCI website • Prepare and submit final GSoC evaluation
Final Evaluation [1–8 September, 2026]		

Deliverables

- A GitHub repository with modular, documented code covering all pipeline stages: data loading, model training, domain adaptation, inference at scale, and evaluation
- Benchmarked baseline models (ENN, EfficientNet-B3, ResNet18) with full training scripts and evaluation notebooks reporting both AUC - ROC and TPR
- Complete domain adaptation implementations (ADDA, WDGRL, SDA) with comparison tables that directly extend the Parul et al. (2024) benchmark
- A strong lens candidate catalogue from HSC-SSP PDR2 Wide data with photometric characterisation and false positive analysis
- Jupyter notebooks reproducing all key experiments: baseline benchmarking, domain adaptation evaluation, inference pipeline, and candidate characterisation
- A comprehensive README with setup instructions, dataset access guide, and a summary of findings
- A short blog post for the ML4SCI website summarising the project results

Future Scope

The most natural extension of this project is to integrate the lens-finding pipeline with downstream dark matter substructure classification (as in Common Test I). An end-to-end pipeline that first detects lenses and then characterises their dark matter content - distinguishing WIMP-like spherical substructure from superfluid vortex signatures - would be a significant contribution to the DeepLense ecosystem.

Beyond astronomy, the domain adaptation and extreme class imbalance techniques developed here apply directly to other rare event searches in high-energy physics, medical imaging, and industrial anomaly detection.

The Motivation for GSoC

I got into machine learning by building everything from scratch. My first serious project was a scalar autograd engine - implementing the Value class, the dynamic computation graph, and backpropagation by hand - and then validating every gradient against PyTorch. It was slow, sometimes frustrating, and completely worth it. I have a genuine feel for what's happening inside these models because I built the machinery before I used the libraries.

What draws me specifically to this project is that it is not a trivial problem. The lens-finding task has a direct scientific consequence - more detected lenses means more probes of dark matter substructure and cosmological parameters. Working through Task V already gave me a taste of that: the 100:1 class imbalance is worth the practice. Diagnosing that, and building a solution that addresses the root cause rather than papering over it, was the most interesting part of the test.

I am also genuinely excited about working with these specific mentors. Anna Parul was the first author on the NeurIPS 2024 domain adaptation paper that serves as one of the two primary references for this project. Sergei Gleyzer and Michael Toomey wrote the 2019 substructure classification paper that motivated the entire DeepLense direction. Working directly with the people who built the foundation of this field is not something I take lightly.

Contributing to open-source scientific infrastructure that will actually be used in real life, that is a strong motivation for me, beyond any credential or stipend.

Achievements

Hackathons and Competitions

1. **1st Place - ML Track, PhysisTechne Symposium Hackathon, IIT (ISM) Dhanbad**

I built QSetup - an autonomous AI for designing photonic quantum circuits. The system uses a genetic algorithm (selection, crossover, mutation, elitism) with quantum state fidelity scoring via QuTiP simulation, and TF-IDF retrieval over research papers to ground generated designs in the literature. Delivered as a full-stack Flask + Next.js application with interactive visualisation.

2. **2nd Place - AI Summit 2026 (60–80 participating teams)**

Autonomous job-search and auto-apply agent: the system ingests a resume, GitHub, and LinkedIn profile; semantically matches them to job listings; generates grounded cover letters via the Gemini API; and auto-submits with safety gates. Team project - I contributed to AI design, prompt engineering, and system architecture. Built with FastAPI and Next.js.

3. **3rd Place - CyberLabs WOC Kaggle Competition (Private Leaderboard, score 0.66725)**

Binary classification on a 3,072-feature dataset with 10% class imbalance. I iterated through 15+ approaches including PCA (95% variance threshold), stratified K-fold out-of-fold cross-validation, decision threshold optimisation (balanced accuracy 0.52 → 0.78), and probability calibration. Key finding: calibrated logistic regression with PCA preprocessing outperformed deep MLPs under strict cross-validation.

4. **Currently competing - LC × MNNIT AI Hackathon on Kaggle**

I am actively competing in this live Kaggle hackathon alongside the GSoC submission period.

5. **AIR 139 - WBJEE | 95% CBSE Class 12 | 96% CBSE Class 10**

Relevant Projects

- **Micrograd + Character-Level Language Model:**

  Scalar autograd engine from scratch with dynamic compute graph and manual backpropagation, validated against PyTorch. Extended to a character-level name generator MLP with learned 2D embeddings, batch normalisation, Kaiming initialisation, and 100k-step mini-batch SGD.

- **ML Library from Scratch (CyberLabs WOC):**

  Linear, Polynomial, and Logistic Regression; KNN (macro-F1 0.742); K-Means; feedforward networks (96.3% validation accuracy) - all in NumPy.

- **EEG Hyperscanning Analysis (NeuroDyads GSoC pre-task):**

  Preprocessed simultaneous 64-channel dyadic EEG (bandpass filtering, bad-channel interpolation via pyprep, ICA artifact removal). Trained CEBRA on joint speaker-listener embeddings; achieved 98.35% KNN decoding accuracy versus 49.45% shuffled baseline.

Certifications

- Convolutional Neural Networks - DeepLearning.AI / Andrew Ng (March 2026)
- Machine Learning Specialization - DeepLearning.AI / Stanford Online (December 2025)

Thank You